

## EXPERIMENT 28

### PROLOG PROGRAM – MEDICAL DIAGNOSIS SYSTEM

#### Aim

To implement a simple **medical diagnosis system using Prolog** that identifies a possible disease based on the symptoms provided by the user.

#### Algorithm

1. Define diseases and their corresponding symptoms.
2. Ask the user to enter symptoms.
3. Compare the entered symptoms with the disease rules.
4. If the required symptoms match, identify the possible disease.
5. Display the diagnosis.
6. If no disease matches, display that the disease could not be identified.

#### Prolog Code

```
% Medical Diagnosis System
```

```
diagnose(S1, S2, S3) :-
```

```
    disease(S1, S2, S3, Disease),
```

```
    write('Possible Disease: '),
```

```
    write(Disease), nl.
```

```
% Disease rules
```

disease(fever, cough, headache, flu).

disease(cough, runny\_nose, sneezing, common\_cold).

disease(fever, chills, sweating, malaria).

disease(fever, abdominal\_pain, weakness, typhoid).

disease(\_, \_, \_, unknown).

### Input

diagnose(fever, cough, headache).

### Output

```
| ?- consult('C:/Users/MOSHIYA/Downloads/exp 28.pl').  
compiling C:/Users/MOSHIYA/Downloads/exp 28.pl for byte code...  
C:/Users/MOSHIYA/Downloads/exp 28.pl compiled, 16 lines read - 1309 bytes written, 18 ms  
  
(16 ms) yes  
| ?- diagnose(fever, cough, headache).  
Possible Disease: flu  
  
true ?
```

### Result

Thus, the medical diagnosis system was successfully implemented in Prolog, and the possible disease was identified based on the symptoms given as input.